ALGORITHM ARENA

[Problem Title]

Formal Problem Specification

Document A: Domain and Mathematical Specification

Problem Identifier:	[problem-slug]
Specification Level:	Document A (Pure Mathematical Problem)
Version:	1.0.0
Category:	Game Theory / Optimization / Control
Difficulty:	Beginner / Intermediate / Advanced / Research
Author:	[Author Name / Team]
Date:	August 20, 2026

Standard Document A template specifying the pure mathematical model, environment state, observation space, action space, transition dynamics, and reward function of a problem.

Contents

1 Metadata

1.1 Problem Name

[Problem Title]

1.2 Short Identifier

`[problem-slug]`

1.3 Category

- ☐ Game Theory
- ☐ Multi-Agent Systems
- ☐ Online Decision Making
- ☐ Combinatorial Optimization
- ☐ Continuous Control / Robotics
- ☐ Reinforcement Learning

1.4 Difficulty Level

- ☐ Beginner
- ☐ Intermediate
- ☐ Advanced
- ☐ Research

1.5 Version History

Version	Date	Author	Changelog Summary
1.0.0	YYYY-MM-DD	Author Name	Initial formalized problem specification

2 Problem Overview

2.1 Description

High-level conceptual description of the system or game being modeled.

2.2 Motivation

Why this problem is valuable, challenging, and interesting from an algorithmic perspective.

2.3 Learning Objectives

Key technical insights, trade-offs, and skills learned by solving this problem.

3 Formal Problem Definition

3.1 Agents and Roles

Number of agents N, symmetry, roles, and capabilities.

3.2 Horizon and Timing

Step structure, episode duration T, and synchronous vs. sequential interaction.

3.3 Action Space

Definition of action set $\mathcal{A}$:

$$\mathcal{A} = \{a_1, a_2, \ldots, a_K\} \quad \text{or} \quad \mathcal{A} \subseteq \mathbb{R}^d$$

3.4 State and Observation Space

State space $\mathcal{S}$ and observation mapping $o_t = \Omega(s_t, h_t)$.

3.5 Transition Dynamics

State transition law: $s_{t+1} \sim \mathcal{P}(s_{t+1} \mid s_t, \mathbf{a}_t)$.

3.6 Reward and Objective Function

Stage reward function $r_t = R(s_t, \mathbf{a}_t, s_{t+1})$ and cumulative utility objective $U(\pi)$.

4 Mathematical Model

4.1 Notation Summary

Symbol	Domain	Meaning
t	$\{0, \ldots, T-1\}$	Time step index
s_t	$\mathcal{S}$	Environment state
a_t	$\mathcal{A}$	Action selected
r_t	$\mathbb{R}$	Stage reward

4.2 Theoretical Properties and Optimality

Known equilibria, optimal bounds, or theoretical constraints.

5 Concrete Execution Examples

Walkthrough of sample steps and transitions.